RESEARCH PARTNERSHIP PROPOSAL / 14 OCTOBER 2026

Bayesian Relativities × Mobility Insurance Profit Testing

A worked, auditable KES illustration for mobility fleets, insurance partners and driver outcomes

Illustrative synthetic case · 100 vehicles · 12 months

Prepared from the Bayesian Relativities paper and its reproducible research case

PARTNERSHIP QUESTION

Can risk, uptime and driver cash improve together?

The proposed work joins actuarial evidence with vehicle operations and driver economics.

01 **Risk**

How can exposure, claims and vehicle-use data produce credible estimates of expected loss with useful uncertainty?

02 **Intervention**

Which combination of insurance design, safety support, repairs and incentives changes outcomes over time?

03 **Profit test**

How do claim development, reinsurance, capital, operating costs and customer cash shape full-term value?

Moove’s public mobility context motivates a test

Public product and CGAP material motivate the question; the case uses synthetic KES assumptions.

Public product context

Moove UK describes vehicle access with weekly payments and packages that include insurance and servicing. Market, contract and payer determine the actual economics.

Research motivation

A 2023 CGAP study discusses use of gig-work data in expanding access to credit and includes Moove. That motivates careful evidence design for mobile work.

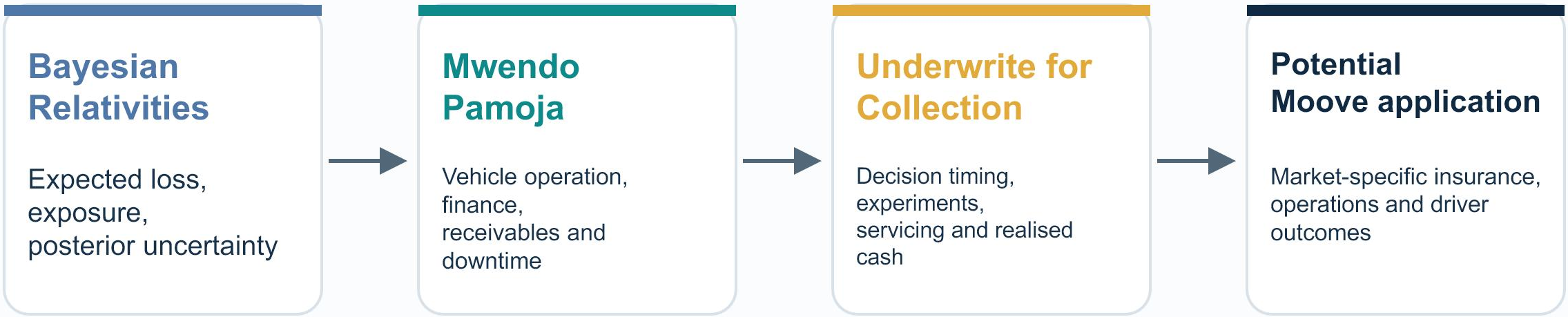
What is proposed: test whether exposure-normalised insurance estimates, service reliability and driver outcomes can be assessed together using a jointly defined market and dataset.

UK public example | Kenya KES case | Synthetic KES illustration

Sources: CGAP [1]; Moove UK Customer Hub [2]; Moove UK Privacy Policy [3].

Three workstreams connect into a Moove test

Each workstream supplies a distinct analytical input.



Explicit data definitions, decision rights and outcome measures connect model evidence to approved market decisions and customer terms.

Price expected loss from exposure, frequency and severity

Keep exposure, underlying expected loss and commercial premium visible as separate quantities.

$$EL_{it} = E_{it} \times \lambda_0 \times \theta_{it} \times \mu_0 \times \omega_{it} + L_{it}^{(non\text{-}drive)}$$

Exposure
km / covered days

→

Frequency
claims per unit

→

Severity
cost given claim

→

Expected loss
risk estimate

Commercial premium = expected loss + claim / operating expense + reinsurance + capital + tax + approved margin

q_i is an exposure-normalised risk relativity; the insurer applies it within approved premium terms and profit tests.

Partial pooling supports credible risk estimates

Cohort experience informs individual estimates as relevant exposure and mature claims accumulate.

Frequency model

$$N_i \mid \theta_i \sim \text{Poisson}(E_i \lambda_o \theta_i)$$

$$\theta_i \sim \text{Gamma}(a, b)$$

$$\theta_i \mid N_i, E_i \sim \text{Gamma}(a + N_i, b + E_i \lambda_o)$$

- Severity is modelled separately, then combined with frequency draws.
- Group structure enables partial pooling across driver, vehicle, territory and time.
- Review outputs include posterior intervals and future-claim predictive ranges.

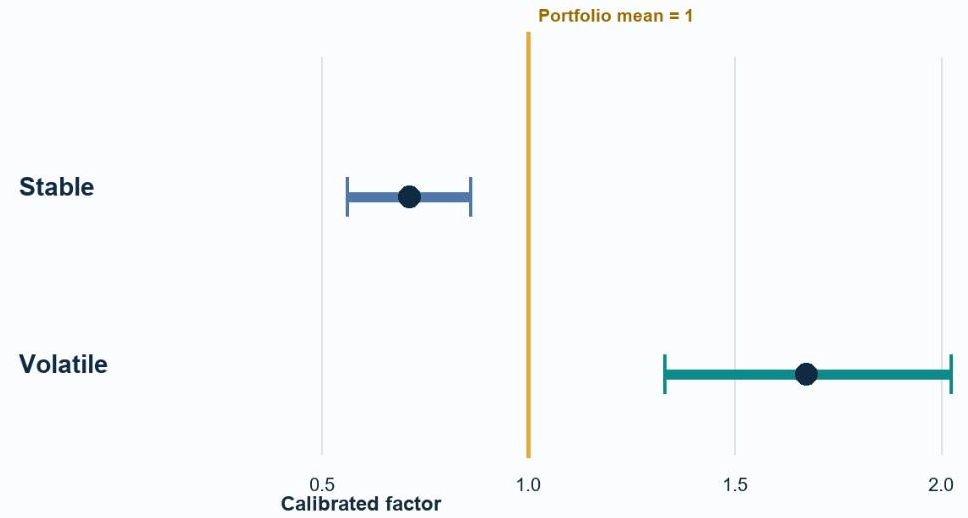
Useful outputs: calibrated relativities · exposure support · posterior uncertainty · predictive loss · refresh lineage

Credible intervals show group risk and uncertainty

Draw-wise calibrated joint expected-loss relativities · synthetic 100-vehicle case

Posterior risk factors retain material uncertainty

Draw-wise calibrated joint expected-loss relativities · 90% credible intervals



Illustrative synthetic KES case · Partner data would calibrate future estimates

Stable cohort

0.711 factor
90% CrI: 0.561–0.859

Volatile cohort

1.674 factor
90% CrI: 1.329–2.023

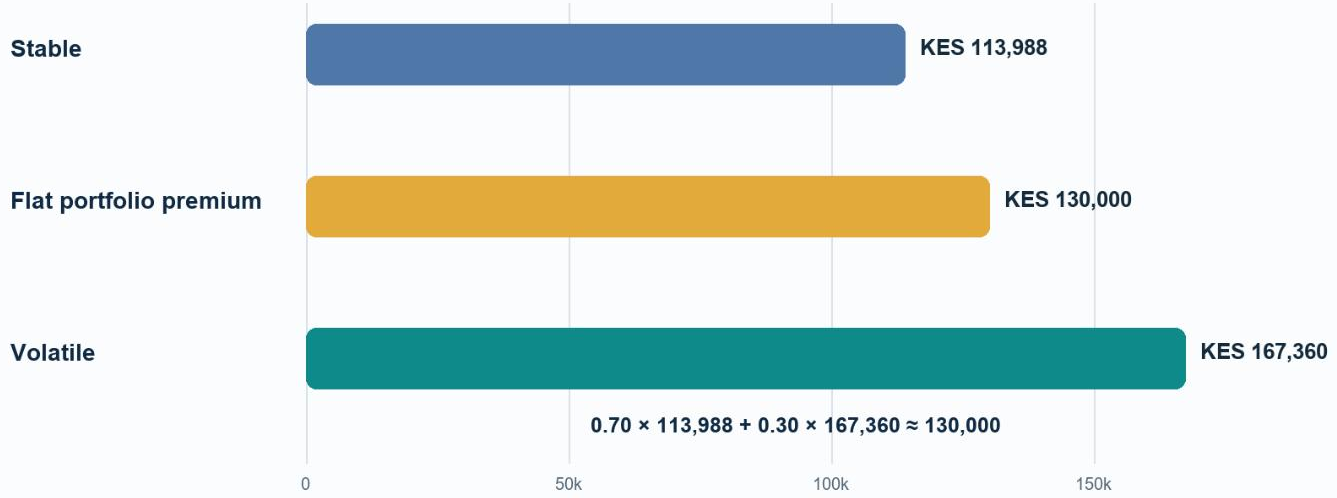
The intervals quantify model uncertainty; future claims also vary around expected loss.

Neutral rates change allocation, not total loss

Exposure-weighted average stays KES 130,000; the commercial layer is held constant in this comparison.

Neutral differentiation reallocates the KES 130,000 premium

Exposure-weighted expected-loss allocation · 70% stable / 30% volatile portfolio



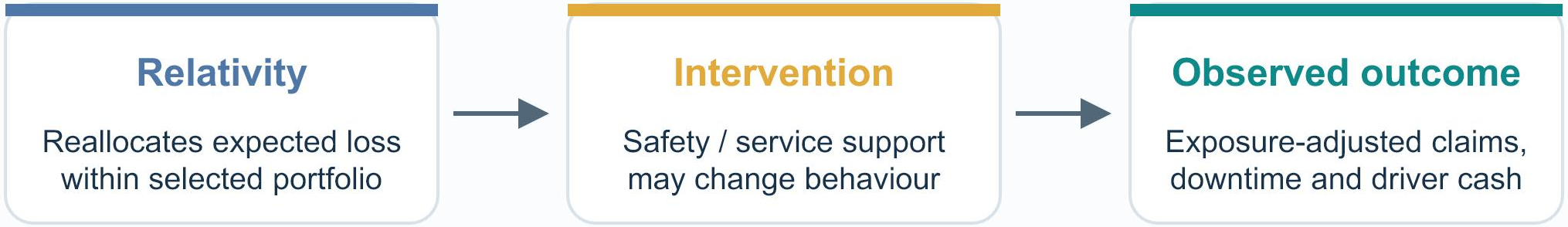
Illustrative synthetic KES case · Partner data would calibrate future estimates

- Stable indicated premium: KES 113,988
- Volatile indicated premium: KES 167,360
- 70% × stable + 30% × volatile = KES 130,000

Risk-tier prices are illustrative allocations. Contract terms, taxes and expenses would need market-specific calibration.

Treat safety and service effects as hypotheses to test

Separate a pricing allocation from an action that is expected to change claim frequency.



Sensitivity input: 10% lower volatile-group frequency. If validated, 75% of the modelled loss saving reduces premium and 25% funds an operator-paid driver incentive.

This effect is assumed for analysis. A pilot with mature, comparable claims estimates the treatment effect.

Profit testing needs three separate ledgers

Each party carries a different contractual cash flow, risk and decision horizon.

INSURER

Premium · claims · expenses
Reinsurance · reserves · capital
Investment cash · shareholder return

MOBILITY OPERATOR

Insurance cost · retained loss
Repairs · downtime · funding
Package, collections · retention

DRIVER

Fares · energy · package
Insurance / excess · incentives
Working days · take-home cash

ΔCF^{M}_{t} = insurance savings + avoided retained losses + uptime / retention value − incentives − intervention / funding costs

Synthetic KES case: 100 vehicles, 12 months

Selected model assumptions are explicit and reproducible.

Portfolio

100 vehicles · 12 months · 70% stable / 30% volatile

History

3 synthetic years · 756 / 324 exposure units of 10,000 km

Claims

47 stable / 41 volatile · mean severities KES 170k / 205k

Posterior

12,000 draws · Gamma–Poisson + inverse-Gamma/Gamma severity

Projection

8,000 predictive paths · 20% quota share · fixed seed 20261014

Shared shock

2% probability; frequency ×1.8 and severity ×1.25 when triggered

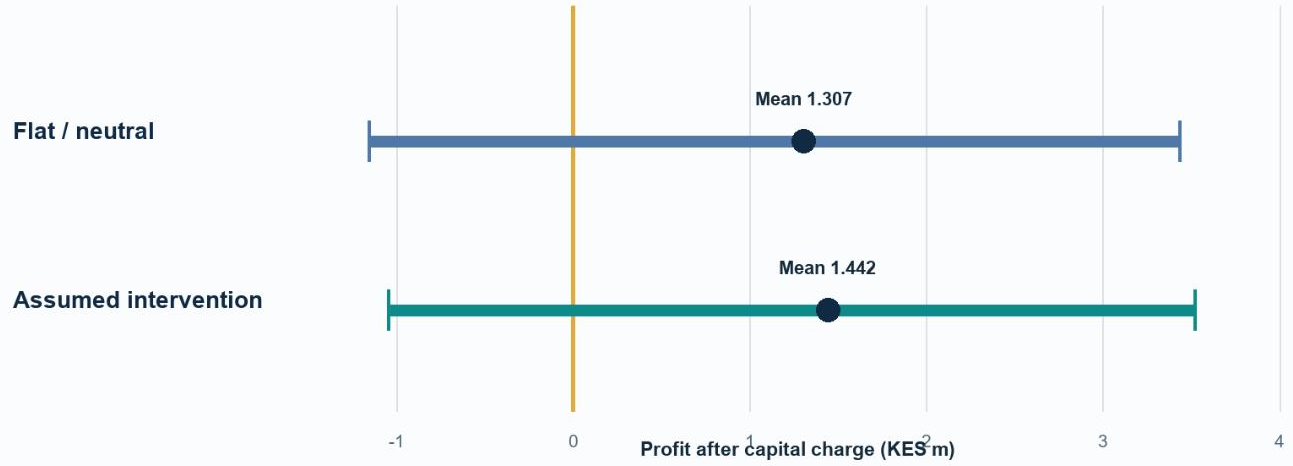
Partner exposure, claims and contract data would calibrate a future application.

Intervention improves mean profit; risk remains

Mean after-capital profit and predictive 90% range · 100-vehicle annual portfolio

Intervention sensitivity improves the mean; downside remains

Insurer profit after a 10% volatile-group frequency reduction assumption · KES millions



Predictive 90% intervals · 8,000 paths · Assumed intervention sensitivity; pilot estimates its effect

Flat / neutral

Mean KES 1.307m
90% range −1.158m to 3.434m

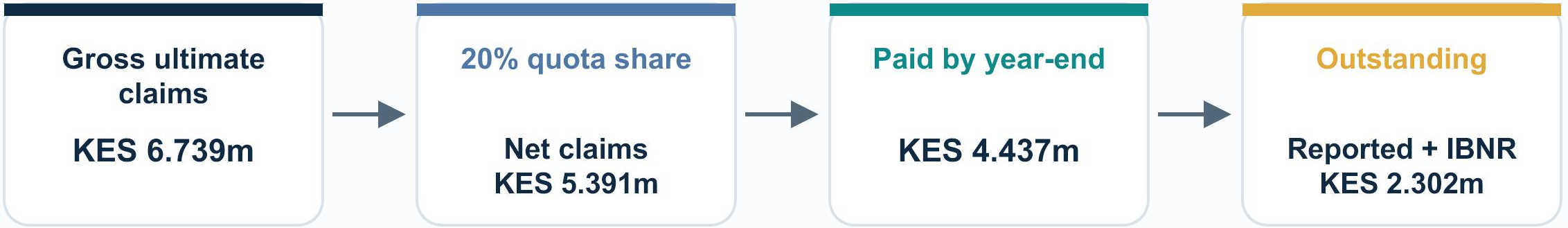
Assumed intervention

Mean KES 1.442m
90% range −1.048m to 3.518m

Probability of loss after capital: 15.8% → 13.4%
Break-even gross premium: KES 107,852 → 103,473 / vehicle-year

Claims, reserves and reinsurance run on separate cash calendars

Annual insurer baseline · gross ultimate claims KES 6.739m



Ceded recoverable at year-end: KES 0.683m · report lag: 1–3 months · settlement: 1–9 months (assumptions)

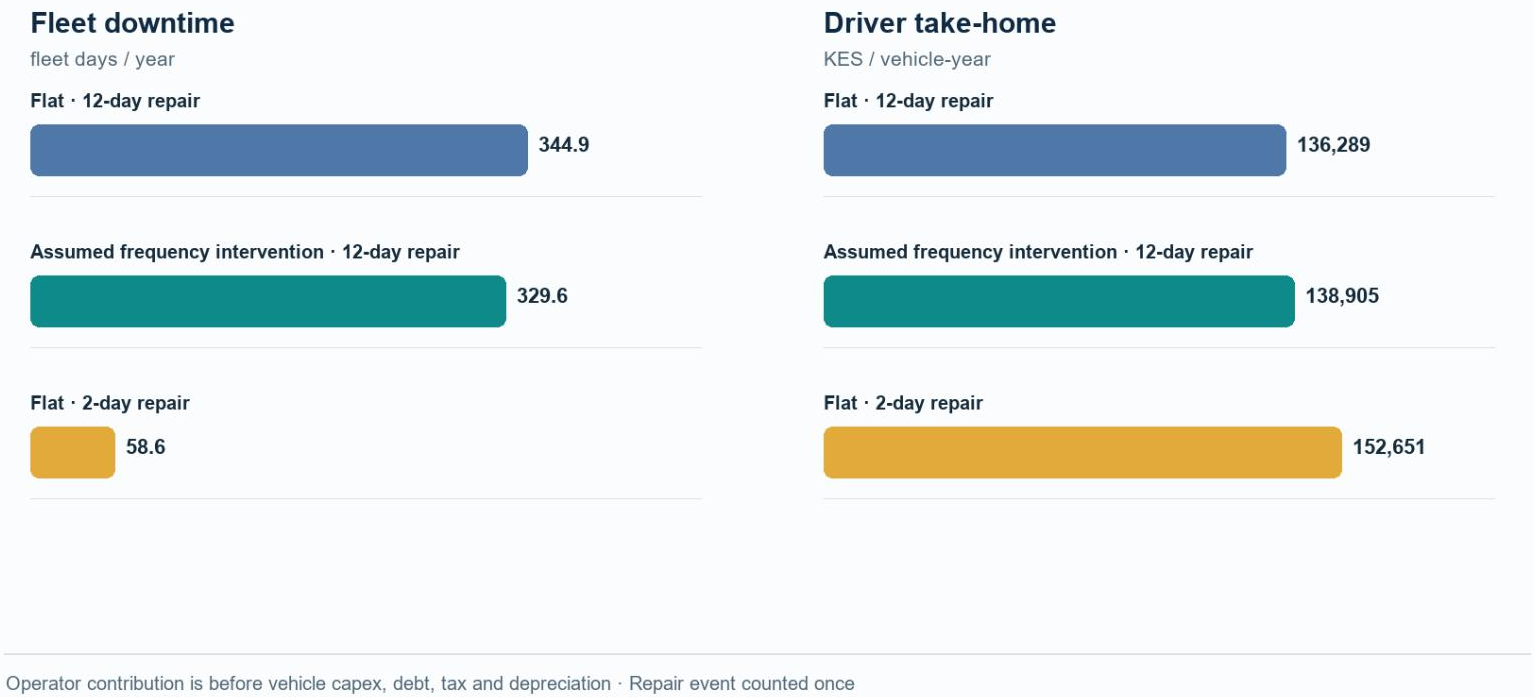
Premium, incurred claims, paid claims, reserve movement, reinsurance recovery and available cash should be projected by period and reconciled.

Repairs change fleet uptime and driver cash

Annual portfolio case · downtime is fleet-wide; take-home is per vehicle-year.

Vehicle availability and driver cash move together

Three separate service assumptions · annual fleet downtime and per-vehicle driver take-home



Operator contribution is before vehicle capex, debt, tax and depreciation · Repair event counted once

Standard service

Operator contribution
KES 1.026m / vehicle-year

Two-day repair

Driver take-home rises to
KES 152,651 / vehicle-year

Expedite cash: KES 146,871 / fleet. Operator contribution excludes vehicle capex, debt, tax and depreciation.

The 99.5% tail is a coarse illustration at 8,000 paths

Treat the capital lens as a scenario result; precision and calibration need further work.

$VaR_{99.5}$

KES 13.093m flat / neutral

$TVaR_{99.5}$

KES 14.491m flat / neutral

Economic capital

KES 7.702m VaR − mean net claims

Tail support

≈ 40 paths upper 0.5% of 8,000

Next sensitivity runs: common-shock size and frequency · reporting / settlement lag · treaty share and recovery delay · capital charge · claims maturity

8,000 paths leave only about 40 observations beyond the 99.5th percentile. Increase runs and assess convergence before using tail values for decisions.

Measure the treatment effect with a pre-specified pilot

Start with the claims and operational data needed to support a fair comparison.

1 Define

Eligible drivers, treatment, standard service and exposure

2 Assign

Randomise or cluster by depot; maintain the agreed baseline

3 Measure

Claims / distance, severity, downtime, earnings, retention

4 Review

Intention-to-treat effect, uncertainty and maturity

Pre-register endpoints, claim maturity windows, exclusions, subgroup analysis and benefit-sharing rules; report costs and customer outcomes with insurer results.

A data and decision contract makes the pilot traceable

Agree data purpose, integration, decision rights and review before connecting systems.

Evidence contract	Exposure units · claim/event dates · repair milestones · payment / reserve basis
Governance	Purpose and access · minimisation / pseudonymisation · retention · model version
Traceability	Correlation ID · event timestamp · decision record · approver · action / outcome
Interfaces	Insurer claims / policy · fleet servicing · driver package / collections · analytics

Use controlled market-specific data access and a documented human review route for exceptions.

A focused first phase can test the partnership

The first deliverable is a jointly agreed, market-specific data and economics blueprint.

01 Select the market, vehicle product and underwriting / claims partner.

02 Agree anonymised, exposure-linked claims, telematics-quality and repair fields.

03 Rebuild baseline rates, reserves and cash flows using mature partner experience.

04 Run a shadow validation; jointly decide whether a controlled pilot is justified.

Research collaboration proposal · No Moove data or performance claims assumed

Sources and reproducibility

Public context, actuarial literature, standards and research-case artefacts.

[1] Anand & Murthy, CGAP (2023), Putting Gig Data to Work.

[2] Moove UK Customer Hub · uk.moove.io/customer-hub/

[3] Moove UK Privacy Policy · uk.moove.io/privacy-policy/

[4] Lee & Shi (2019), Insurance: Mathematics and Economics, doi:10.1016/j.insmatheco.2019.04.004.

[5] Maloba, Bayesian Credibility and Exposure-Normalised Telematics Relativities (2026).

[6] Belhadj, Goulet & Ouellet (2009), ASTIN Bulletin 39(2).

[7] Bühlmann (1967), ASTIN Bulletin 4(3).

[8] Actuarial Standards Board, ASOP 43: Property/Casualty Unpaid Claim Estimates.

[9] IFRS Foundation, IFRS 17 Pocket Guide: Reinsurance Contracts Held.

[10] IAIS, Insurance Core Principles and ComFrame (2024).

[11] Robbin & Malhotra (2023), CAS Forum, capital management and reinsurance profit testing.

Reproducible case: research_lab/mobility_insurance_profit_case/ (seed 20261014).

Full IEEE references, URLs, model assumptions and validation checks are in SOURCE_AND_VALIDATION_NOTES.md.